

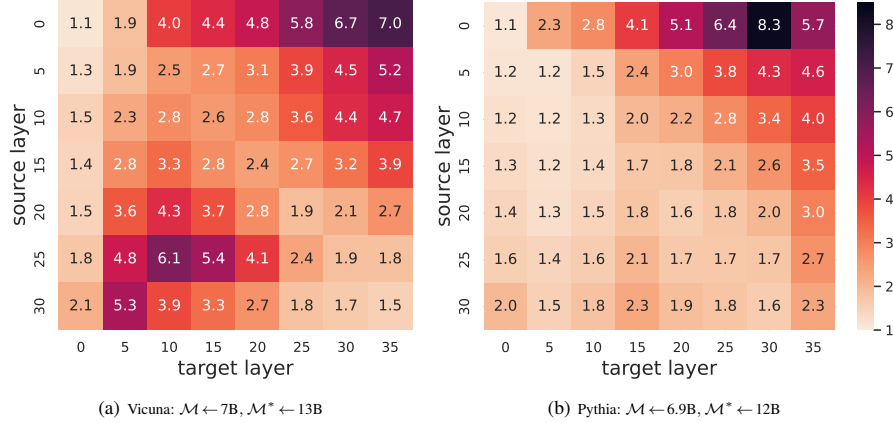


Figure 11. Surprisal scores ($\uparrow$ is better) on next-token prediction estimation in Vicuna and Pythia cross-model Patchscopes.

E. Additional Results on the Cross-Model Patching Experiment

Fig. 10 depicts the Precision@1 scores for different combinations of source-target layers, showing that patching with a simple affine Patchscope proves to be effective with precision of up to 0.7 and 0.8 for Vicuna and Pythia, respectively. Specifically, patching representations to an early layer of the larger model seems to be the most effective. Furthermore, it appears that there is a subtle matching among some layers of the models, with the diagonal consistently exhibiting higher values. As shown in Fig. 11, similar trends are observed in the surprisal results for both models. Overall, these results show that, when $\mathcal{M}^*$ and $\mathcal{M}$ are from the same model family, it is possible to leverage $\mathcal{M}^*$ for decoding information from the representations of $\mathcal{M}$. Notably, our findings extend observations by Csiszárík et al. (2021) from patching representations across deep convolutional neural networks with the same architecture but different initializations.

F. More Details on the Multi-Hop Reasoning Experiment

Data Building on Hernandez et al. (2023b), we systematically generate all valid multi-hop factual and commonsense reasoning queries where the object ω_1 in the first triplet τ_1 is equal to the subject σ_2 in the second triplet τ_2 . This process yields 1,104 multi-hop reasoning samples. We then filter examples to the ones where the model accurately represents both triplets independently. Tab. 8 summarizes information about the samples where Vicuna (13B) correctly represents each reasoning step. Out of 1,104 samples that require two steps of commonsense or factual reasoning, 46 satisfy the above criteria, with 8 unique relation combinations.

More Detailed Results Fig. 12 shows the interaction between ℓ and ℓ^* and how it affects the success rate. Patching representations from most source layers ℓ into early-to-mid ℓ^* (6-16) is the most effective in making the right prediction. Our interpretation is that patching into late ℓ^* is not effective because ρ_2 has already been processed and the result has been copied to the last position, therefore it cannot incorporate the proxy of σ_2 . We also observe that $\ell^* \leq \ell$ is more successful on average.

G. Compute Resources

All experiments were conducted using A100 80GB GPUs, with the exception of experiments with GPT-J (Wang & Komatsuzaki, 2021) where we used A100 40GB GPUs.

Table 6. Additional qualitative examples illustrating entity resolution in **Vicuna 13B**. The expressive generations show that as we go through the layers, more tokens from the context get integrated into the current representation, referred to as “Tokens Covered”. Here, $\mathcal{M} \leftarrow \text{Vicuna 13B}$, $\mathcal{M}^* \leftarrow \mathcal{M}$, $\ell^* \leftarrow \ell$.

S	Tokens Covered	ℓ	Generation	Notes/Entity Described by the Generation
"Will Smith"	"Smith"	1	sonian: National museum and research complex in Washington, D.C.	Smithsonian Museum
	"Smith"	2	sonas: American rock band	An amalgam of Smith band and SONA band
	"Will Smith"	3	: American actor and rapper	Will Smith
"Queen Victoria"	"Victoria"	1	s Secret: American multinational women's lingerie, beauty and body-care products company	Victoria's Secret
	"Victoria"	2	s Secret: American lingerie and beauty products retailer	Victoria's Secret
	"Queen Victoria"	3	: Queen of the United Kingdom of Great Britain and Ireland from 1837 to 1901	Queen Victoria
	"Queen Victoria"	4-8	: Queen of the United Kingdom of Great Britain and Ireland from 1837 until her death in 1901	Queen Victoria
	"Queen Victoria"	9	: Queen of England (reigned from 1837-1901)	Queen Victoria
	"Queen Victoria"	10	: 19th century British queen who ruled from 1837-1901	Queen Victoria
"Paris Hilton"	"ton"	1	y: a type of pasta	Rigatoni/Rigatony/Rigatoné
	"Hilton"	2	Hotels: American multinational hospitality company	Hilton Hotel
	"Hilton"	3	: Hotel chain founded by Conrad Hilton	Hilton Hotel
	"Hilton"	4	: Hotel and resorts company	Hilton Hotel
	"Paris Hilton"	5-6	: Socialite and television personality	Paris Hilton
	N/A	7	: Placeholder for a variable or concept	Placeholder contamination
	N/A	8-9	: Variable representing any number of things or concepts	Placeholder contamination
	N/A	10	: x is a placeholder	Placeholder contamination
"James Bond"	"Bond"	1	: Unit of measurement for the strength of a liquid	Bond Unit
	"James Bond"	2	: James Bond, the fictional British secret agent created by novelist Ian Fleming and portrayed by actor Daniel Craig in the 2006 film "Casino Royale,"	James Bond
	"James Bond"	3-5	: Fictional British secret agent	James Bond
	"James Bond"	6	: Code name for fictional British secret agent James Bond	James Bond
	"James Bond"	7	: Code name for a fictional British secret agent created by novelist Ian Fleming	James Bond
	"James Bond"	8	: Subject of a fictional British secret agent, created by novelist Ian Fleming and portrayed in a series of films.	James Bond
	James Bond	9	: Subject of the 007 novels and films	James Bond
	N/A	10	: Variable representing the number of the film in the James Bond series	Placeholder contamination
"Dwayne Johnson"	"Johnson"	1-2	& Johnson: American multinational corporation in the fields of pharmaceuticals, medical devices, and consumer packaged goods	Johnson & Johnson
	"Dwayne Johnson"	3-4	: American actor and professional wrestler, also known as "The Rock"	Dwayne Johnson
	"Dwayne Johnson"	5	: Also known as "The Rock," American actor and professional wrestler	Dwayne Johnson
	"Dwayne Johnson"	6	: Former professional wrestler and actor	Dwayne Johnson
	N/A	7	: Abbreviation for the United States	Placeholder contamination
	N/A	8	: Variable representing any number of things or concepts	Placeholder contamination
	N/A	9	: Variable representing any number of other options.	Placeholder contamination
	N/A	10	: Variable representing any number of other options or possibilities.	Placeholder contamination

Table 7. Additional qualitative examples illustrating entity resolution in **Pythia 12B**. The expressive generations show that as we go through the layers, more tokens from the context get integrated into the current representation, referred to as “Tokens Covered”. Here, $\mathcal{M} \leftarrow \text{Pythia 12B}$, $\mathcal{M}^* \leftarrow \mathcal{M}$, $\ell^* \leftarrow \ell$.

S	Tokens Covered	ℓ	Generation	Notes/Entity Described by the Generation
"Will Smith"	"Smith"	1	& Wesson: American firearms manufacturer	Smith & Wesson
	"Will Smith"	2	: American actor	Will Smith
"Alexander the Great"	"Great"	1	Britain: Country in the European Union	Great Britain
	"the Great"	2	Wall Street Crash of 1929: Financial crisis in the United States	the Great Depression
	"the Great"	3	Wall Street Bubble: The Great Depression	the Great Depression
	"the Great"	4	Wall Street: Wall Street in New York City	Wall Street (related to the Great Depression)
	"Alexander the Great"	5	: Ancient Greek ruler, and the first to rule all of the then known world	Alexander the Great
"George Washington" $S \leftarrow$ "Alexander the Great"	"Washington"	1	: Capital of the United States	Washington D.C.
	"Washington"	2	: American capital city	Washington D.C.
	"George Washington"	3	: American president	George Washington
	"George Washington"	4-6	: American revolutionary	George Washington
	"George Washington"	7	: American president	George Washington
	"George Washington"	8	: George Washington	George Washington
	"George Washington"	9	: George Washington, the first President of the United States	George Washington
	"George Washington"	10	: George Washington, Thomas Jefferson, Abraham Lincoln, and Theodore Roosevelt: American presidents	George Washington
"Titanic"	"c"	1	loud: Apple’s cloud-based storage service	cloud
	"Titanic"	2	: Sinking ocean liner	Titanic
	"Titanic"	3	: The ship that sank in the Atlantic Ocean in 1912	Titanic
	"Titanic"	4-6	: British passenger ship	Titanic
	"Titanic"	7-9	: The ship that sank	Titanic
	"Titanic"	10	: Titanic, Titanic: British luxury passenger liner	Titanic
"Saturday Night Live"	"Live"	1	Nation: American concert promoter	Live Nation
	"Live"	2	Nation: American television network	Live Nation
	"Saturday Night Live"	3	: American television program	Saturday Night Live
	"Saturday Night Live"	4-10	: American television comedy show	Saturday Night Live
"Nineteen Eighty-Four"	"Four"	1	Seasons: American hotel chain	Four Seasons Hotel
	"Four"	2	Seasons: The four seasons of the year	Four Seasons (of the year)
	"Eighty-Four"	3	Lions: Chinese professional football club	Cangzhou Mighty Lions F.C.
	"Nineteen Eighty-Four"	4	: Number of the novel 1984 by George Orwell	Nineteen Eighty-Four (Novel)
	"Nineteen Eighty-Four"	5	: George Orwell’s novel, and the 1984 film.	Nineteen Eighty-Four (Novel & Film)
	"Nineteen Eighty-Four"	6	: George Orwell’s novel	Nineteen Eighty-Four (Novel)
	"Nineteen Eighty-Four"	7	: George Orwell: British novelist	Nineteen Eighty-Four (Novel)
	"Nineteen Eighty-Four"	8	: 1984 novel by George Orwell	Nineteen Eighty-Four (Novel)
	"Nineteen Eighty-Four"	9-10	: 1984 novel by George Orwell, and the 1984 film adaptation directed by Michael Radford.	Nineteen Eighty-Four (Novel & Film)